

Meta

AI Infrastructure Architecture Evolution

2015-2026

Silicon generations · Memory and interconnects
Conference disclosures · Systems and software

A generation-by-generation synthesis of conference disclosures and primary technical sources.
Evidence cutoff: 28 September 2026.
Complete English edition, paired with a full Simplified Chinese report.

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01 Scope and executive synthesis

This report covers Meta's AI infrastructure from 2015 to 28 September 2026: MTIA accelerators, their integrated AI networking, system packaging and software, plus the related MSVP video-processing branch. Merchant CPUs, GPUs and switch ASICs are distinguished from Meta-designed silicon. Earlier years establish infrastructure context rather than an invented annual sequence of custom chips.

MTIA evolves from low-power recommendation inference into an HBM-based training platform and then an inference-oriented generative-AI roadmap. The decisive transition is not merely more arithmetic: it adds network chiplets, near-memory reduction and rack-scale communication while preserving a PyTorch-facing software contract. Meta's workload ownership allows narrower optimization than a merchant GPU, but portability, reliability and model evolution remain constraints.

02 MTIA100 and MTIA200: recommendation inference

MTIA100, originally MTIA v1, was designed around recommendation inference and disclosed at ISCA 2023. Its 7 nm implementation combines an 8×8 processing-element grid, local memories and a larger shared SRAM tier with LPDDR5. The key architectural balance is enough arithmetic for dense portions of the model without sacrificing the storage and access behavior needed by sparse features. This is a different objective from maximizing large-batch transformer training FLOPS.

MTIA200, originally MTIA2i, retains the 64-PE organization while moving to 5 nm, higher frequency, more local storage and 256 MB shared SRAM. Dense FP16/BF16 rises from 51.2 to 177 TFLOP/s, while LPDDR5 bandwidth increases only from 176 to 204.8 GB/s. The design therefore relies increasingly on on-chip reuse and sparse-workload scheduling. The 2024 platform fits 72 accelerators in a rack, but this is not the same as the 72-device dedicated scale-up domain introduced with MTIA400.

03 MTIA300: training changes the data path

The ISCA 2026 paper describes one compute chiplet, two network chiplets and six HBM3e stacks. A 12×6 active PE array combines RISC-V vector control, dot-product, special-function, reduction and DMA engines. The memory system provides 216 GB and 6.1 TB/s. Redundant PE resources address yield and faults. Moving from LPDDR to HBM changes both packaging and the balance between dense layers, embeddings and collective traffic.

Each network chiplet integrates six 800 Gb/s RDMA controllers. The aggregate endpoint rate is 1.2 TB/s in each direction. Message engines and near-memory reduction let collectives progress without occupying the principal compute engines or passing the payload through the host PCIe link. The paper's deployed scale-up configuration uses 800 GB/s, with support up to 1,000 GB/s; scale-out is 200 GB/s. The marketing table uses the supported maximum. Both configurations are retained rather than silently merged.

A one-to-one host-CPU/accelerator relationship supports optimizer offload. Large HBM capacity can permit larger local batches and fewer communicating trainers, but the benefit depends on optimizer state, precision and convergence constraints. HCCL can fuse communication and computation; the reported collective results are workload- and topology-specific. Integrated NICs reduce one bottleneck, not every synchronization or network bottleneck.

04 MTIA400, 450 and 500: the inference roadmap

The March 2026 roadmap identifies MTIA300 as in production, MTIA400 as laboratory-tested and moving toward deployment, and MTIA450/500 as planned for 2027. MTIA400 uses two compute chiplets and a 72-device scale-up system. MTIA450 prioritizes memory bandwidth and low-precision inference. MTIA500 moves to four compute chiplets plus separate network and host-I/O functions. These are disclosure statuses, not assumed public-cloud or merchant availability.

Architectural interpretation: reusing chassis and network infrastructure can shorten deployment cycles, but it fixes parts of the system bandwidth budget while compute and HBM continue to grow. From 400 to 500, scale-up remains 1.2 TB/s and scale-out 100 GB/s per device, while HBM bandwidth triples. The compiler must consequently preserve more locality or change parallelization. A 25× comparison from MTIA300 MX8 to MTIA500 MX4 crosses numerical formats and is not a like-for-like arithmetic speedup.

05 Generation specifications and disclosure reconciliation

MTIA reference specifications - 1/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Architecture	GA / milestone	Status	Process	Dies / packaging
MTIA100 / v1	Recommendation inference	2023 disclosure	Historical shipping	7 nm	1 logic die
MTIA200 / 2i	Recommendation inference	2024 deployment	Shipping / available	5 nm	1 logic die
MTIA300	R&R training	2026 production	Shipping / available	n/d	1 compute + 2 network; interposer
MTIA400	GenAI / inference	2026 deployment path	Announced; GA n/d	n/d	2 compute chiplets
MTIA450	GenAI / inference	early 2027 target	Roadmap	n/d	n/d
MTIA500	GenAI / inference	2027 target	Roadmap	n/d	4 compute + 2 network + SoC

MTIA reference specifications - 2/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Compute units	Dense BF16	Dense FP8	Dense FP6 / FP4	FP64 vector / matrix
MTIA100 / v1	64 PEs	51.2 TFLOP/s	n/a	n/a	n/d
MTIA200 / 2i	64 PEs	177 TFLOP/s	n/a	n/a	n/d
MTIA300	72 active PEs	560 TFLOP/s (paper)	1120 TFLOP/s (paper)	n/d	n/d
MTIA400	n/d	3 PFLOP/s advertised; dense basis n/d	6 PFLOP/s FP8/MX8; dense basis n/d	12 PFLOP/s MX4; dense basis n/d	n/d
MTIA450	n/d	3.5 PFLOP/s advertised; dense basis n/d	7 PFLOP/s FP8/MX8; dense basis n/d	21 PFLOP/s MX4; dense basis n/d	n/d
MTIA500	n/d	5 PFLOP/s advertised; dense basis n/d	10 PFLOP/s FP8/MX8; dense basis n/d	30 PFLOP/s MX4; dense basis n/d	n/d

MTIA reference specifications - 3/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Memory: type; stacks; GB; TB/s	LLC / SRAM
MTIA100 / v1	LPDDR5; 64 GB; 0.176 TB/s	128 MB; 128 KB/PE
MTIA200 / 2i	LPDDR5; 64-128 GB; 0.2048 TB/s	256 MB; 384 KB/PE
MTIA300	HBM3e; 6; 216 GB; 6.1 TB/s	192 MB; 512 KB/PE
MTIA400	HBM; stacks n/d; 288 GB; 9.2 TB/s	n/d
MTIA450	HBM; stacks n/d; 288 GB; 18.4 TB/s	n/d
MTIA500	HBM; stacks n/d; 384-512 GB; 27.6 TB/s	n/d

MTIA reference specifications - 4/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Scale-up: links; GB/s per direction; domain	Host link	Power / cooling	Form factor
MTIA100 / v1	n/a	PCIe4 x8	25 W	Accelerator board
MTIA200 / 2i	PCIe fabric; not dedicated scale-up	PCIe5 x8	90 W launch; 85 W paper	Accelerator board
MTIA300	800-1000 GB/s one-way; 16 devices	PCIe5 x16	912 W TDP / 667 W typical (paper)	Liquid-cooled module
MTIA400	1200 GB/s one-way; 72 devices	n/d	1200 W module; liquid/AALC	Rack module
MTIA450	1200 GB/s one-way; 72 devices	n/d	1400 W module; liquid/AALC	Rack module
MTIA500	1200 GB/s one-way; 72 devices	n/d	1700 W module; liquid/AALC	Rack module

The March roadmap lists MTIA300 at 0.6 PFLOP/s BF16, 1.2 PFLOP/s FP8/MX8 and 800 W module TDP. The ISCA paper lists 560/1,120 TFLOP/s and 912 W TDP with 667 W typical power. These represent different published configurations or measurement bases; no undocumented reconciliation is imposed. For derived ratios, the paper’s matched values are used. Likewise, MTIA200’s launch states 90 W while the later comparison table states 85 W. Sparse peaks are kept separate from dense values.

Host CPU boundary - 1/3

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Generation / reference SKU	Codename	GA	Status	Core march	Cores / threads
External host CPUs	n/a	n/a	Supplier products	n/a	n/a

Host CPU boundary · 2/3

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Generation / reference SKU	Process per die	Dies / packaging	L2 per core	L3 / domain
External host CPUs	n/a	n/a	n/a	n/a

Host CPU boundary · 3/3

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Generation / reference SKU	Memory: channels; rate; GB/s	Sockets / links	PCIe / CXL	TDP	Vector / matrix ISA
External host CPUs	n/a	n/a	n/a	n/a	n/a

06 Integrated NICs, networking and MSVP

MTIA network chiplets · 1/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	GA / milestone	Status	Class
MTIA300 network chiplets	2026	Shipping / available	Integrated AI NIC

MTIA network chiplets · 2/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Ports × speed	SerDes	Host interface	Programmable engine
MTIA300 network chiplets	2×6×800 Gb/s	n/d	Die-to-die	Message engines / near-memory reduction

MTIA network chiplets · 3/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Embedded CPU	On-card memory	Transport / RDMA	Congestion control
MTIA300 network chiplets	RISC-V control	Shared accelerator memory	RDMA	Credit-based fabric mechanisms

MTIA network chiplets · 4/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Key offloads	Power
MTIA300 network chiplets	Collectives; reductions; direct data path	Within module

MSVP is a dedicated video-transcoding ASIC disclosed in 2023. It belongs in the infrastructure portfolio because video processing consumes substantial fleet resources, but it is not an MTIA generation or a general neural-network accelerator. Likewise, Meta’s OCP server and networking designs, including switch software and rack standards, should not be confused with ownership of every merchant ASIC they contain.

Interconnect evolution · 1/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Interconnect	Year / generation	Lane rate	Lanes / link	GB/s per link per direction
MTIA200 host fabric	2024	PCIe5 32 GT/s	8	~31.5 before packet overhead
MTIA300 scale-up	2026	800 Gb/s port	n/d	100
MTIA400/450/500 scale-up	2026 roadmap	n/d	n/d	1200 aggregate/device

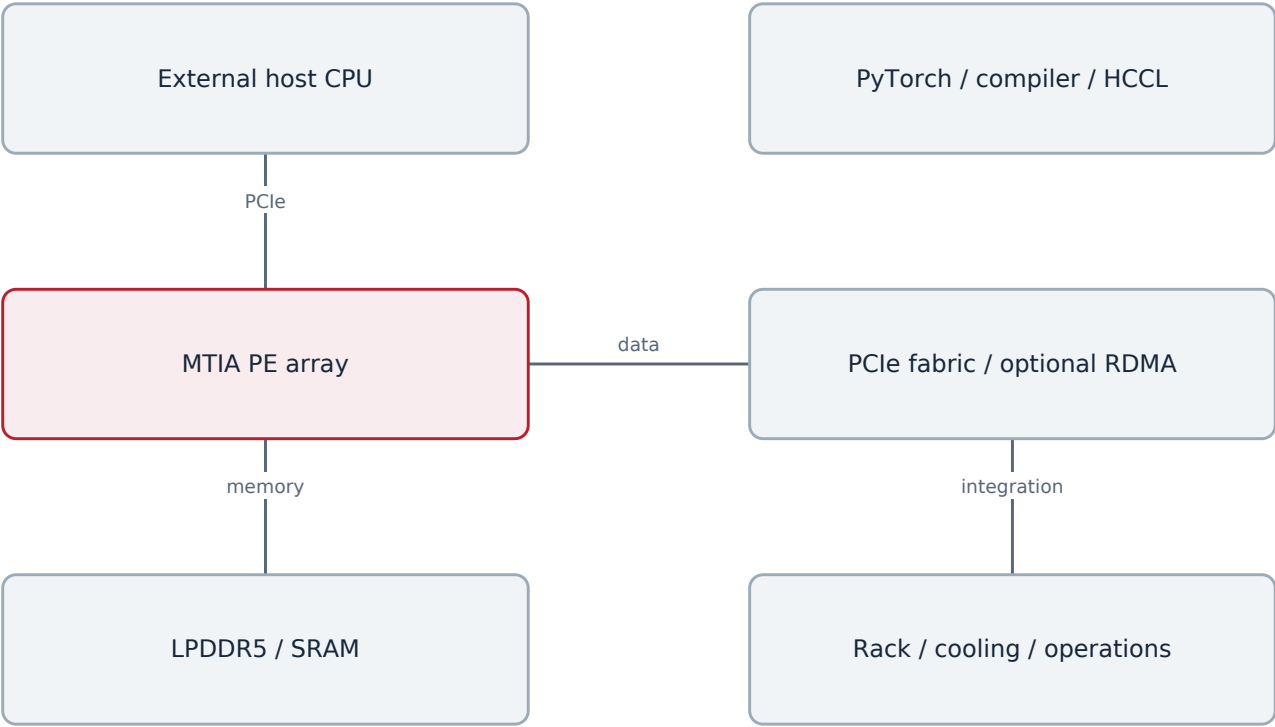
Interconnect evolution · 2/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Interconnect	Links / device	Topology	Domain limit	Coherence / semantics
MTIA200 host fabric	n/d	PCIe switching	72/rack is packaging count	PCIe
MTIA300 scale-up	Configuration-specific	Rack fabric	16	RDMA; explicit collectives
MTIA400/450/500 scale-up	n/d	Switched backplane	72	Explicit communication

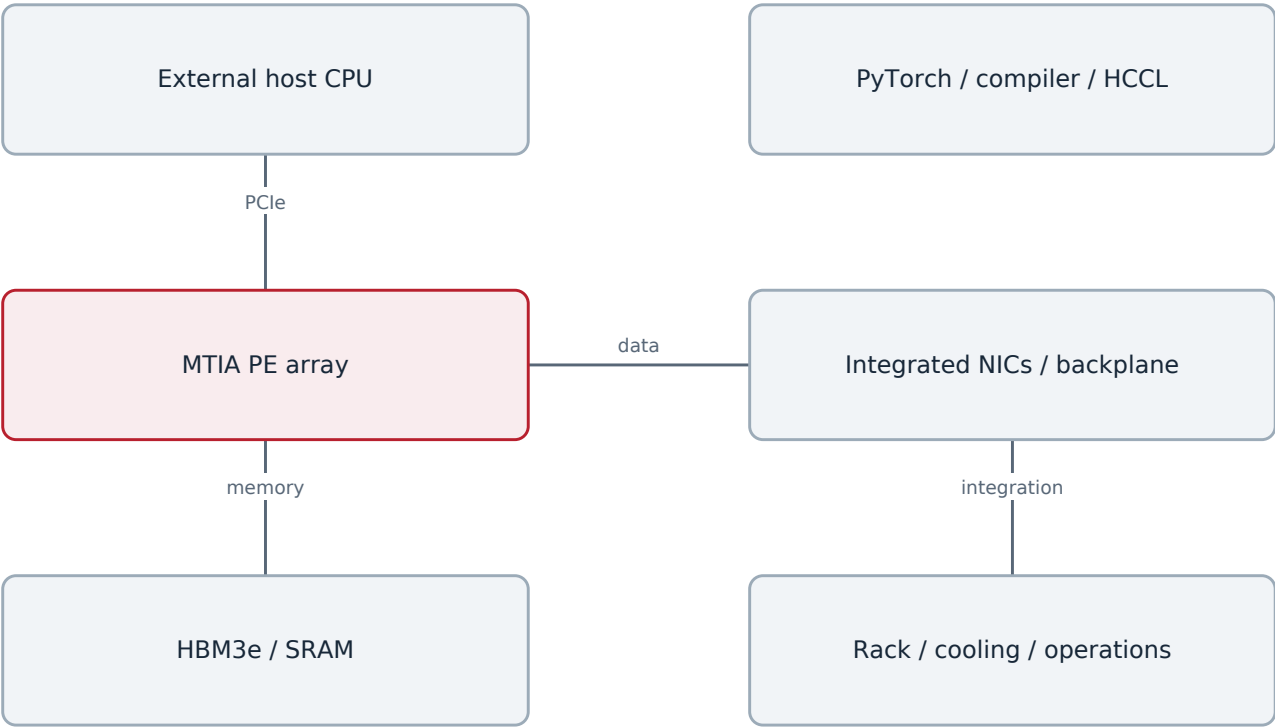
07 Rack architecture and software

2023-2024 stack



Later diagram includes announced products; silicon and rack availability are tracked separately.

2026 roadmap stack



Later diagram includes announced products; silicon and rack availability are tracked separately.

System generations · 1/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Platform	Year / milestone	CPU / accelerator count	Scale-up domain
MTIA200 rack	2024	72 accelerators; 3 chassis	PCIe topology
MTIA300 rack	2026	16 accelerators; 1 CPU/accelerator	16
MTIA400/450/500 rack	2026-2027 targets	72 accelerators	72

System generations · 2/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Platform	Scale-out per accelerator	Rack power	Cooling
MTIA200 rack	Optional external RDMA NIC	n/d	Air
MTIA300 rack	200 GB/s nominal one-way	n/d	Liquid
MTIA400/450/500 rack	100 GB/s nominal one-way	n/d	AALC / facility liquid

Software stack

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Layer	Component	Function	Architectural implication
Frontend	PyTorch / torch.compile / export	Model capture	Shared framework interface
Compiler	TorchInductor / Triton / MLIR / LLVM	Graph and kernel lowering	Schedule local memories and engines
Communication	Hoot Collective Communications Library	Collectives and reduction offload	HCCL here is Meta Hoot
Runtime	Rust user-space driver / firmware	Memory, execution, observability	Host and device coordination
Serving	vLLM plugin	Serving integration	Backend-specific optimized kernels

08 Normalized trends and engineering tradeoffs

Memory and communication balance

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product / metric	Calculation	Interpretation
MTIA100 DRAM/BF16	$176e9/51.2e12 = 0.00344 \text{ B/FLOP}$	Dense peak
MTIA200 DRAM/BF16	$204.8e9/177e12 = 0.00116 \text{ B/FLOP}$	More reuse required than MTIA100
MTIA300 HBM/BF16	$6.1e12/560e12 = 0.0109 \text{ B/FLOP}$	Paper configuration
MTIA300 capacity/BF16	$216e9/560e12 = 3.86e-4 \text{ B}\cdot\text{s/FLOP}$	Per accelerator
MTIA300 scale-up/HBM	$0.8/6.1 = 0.131$	Deployed paper configuration
MTIA400→500 scale-up/HBM	$1.2/9.2=0.130 \rightarrow 1.2/27.6=0.0435$	Network grows slower than HBM
MTIA300 BF16/TDP	$560/912 = 0.614 \text{ TFLOP/s/W}$	Peak/TDP; not application efficiency
Host DDR/core and L3/core	n/a	No Meta host-CPU SKU asserted

The large increase in bytes per BF16 FLOP at MTIA300 reflects a workload choice: recommendation training needs embedding capacity and traffic handling. Subsequent inference products raise both HBM and low-precision compute but retain a fixed rack network budget. Their success depends on model placement, expert routing, KV-cache behavior and compiler utilization. Compare at matched precision, quality, batch size and service-level latency, with module and rack power kept separate.

09 Conference and product crosswalk

Disclosure map · 1/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Venue	Year	Paper / talk / disclosure
ISCA	2023	MTIA v1
ISCA	2025	Second-generation MTIA
ISCA	2026	MTIA300 built-in NICs
AI Infra @Scale	2024	Hardware and co-design of MTIA
SC	2026	HCCL
Company technical disclosure	2026	Four MTIA chips in two years

Disclosure map · 2/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Venue	Presenter / authors	Product	Evidence type	Source
ISCA	Meta authors	MTIA100	Architecture paper; official companion	C02 P03
ISCA	Meta authors	MTIA200 / 2i	Paper record; official architecture	C03 P04
ISCA	Meta authors	MTIA300	Full paper	C01
AI Infra @Scale	Colburn; Lan; Montgomery	MTIA200	Engineering presentation	P02
SC	Meta authors	MTIA communication	Preprint; November event still future	C04
Company technical disclosure	Meta	300 / 400 / 450 / 500	Product and roadmap	E01

No product-specific ISSCC, HPCA, MICRO or ASPLOS circuit/architecture chain is established here for MTIA. ISCA provides the strongest direct product evidence. The public roadmap is broader than the available implementation papers: process, die details and dense/sparse arithmetic basis remain incomplete for 400/450/500. The report retains those gaps in the tables rather than extrapolating earlier silicon details.

Naming and ownership

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Name	Alias / role	Boundary
MTIA100	MTIA v1	Inference ASIC
MTIA200	MTIA2i	Not MTIA300 training chip
HCCL	Hoot	Not Huawei HCCL
MSVP	Video transcoding	Separate ASIC branch
Broadcom	MTIA development partner	Not a merchant MTIA SKU

ISSCC background · 1/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Venue	Year	Paper / talk / disclosure
ISSCC plenary	2019	Deep Learning Hardware: Past, Present, & Future

ISSCC background · 2/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Venue	Presenter / authors	Product	Evidence type	Source
ISSCC plenary	Yann LeCun	Research context; not MTIA implementation	Official plenary record	C05

Source map

- 01 Scope and executive synthesis - C01, E01, E03, E04, P03, P04
- 02 MTIA100 and MTIA200: recommendation inference - C02, C03, P03, P04
- 03 MTIA300: training changes the data path - C01, C04, E01, P01
- 04 MTIA400, 450 and 500: the inference roadmap - E01
- 05 Generation specifications and disclosure reconciliation - C01, E01, E03, P03, P04
- 06 Integrated NICs, networking and MSVP - C01, E01, E03, E04, P01, P04
- 07 Rack architecture and software - C01, C04, E01, P01, P04
- 08 Normalized trends and engineering tradeoffs - C01, E01, P04
- 09 Conference and product crosswalk - C01, C02, C03, C04, C05, E01, E04, P02, P03, P04

References

Product and technical documentation

[P01] MTIA300 built-in NICs and collective offload

<https://engineering.fb.com/2026/08/24/networking-traffic/mtia-300-meta-training-chip-built-in-nics/>

Primary source; accessed by 2026-09-28

[P02] Inside MTIA hardware co-design, 2024

<https://engineering.fb.com/2024/08/22/ml-applications/meta-mtia-hardware-co-design/>

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[P03] First generation MTIA architecture

<https://ai.meta.com/blog/meta-training-inference-accelerator-AI-MTIA/>

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[P04] Second generation MTIA architecture

<https://ai.meta.com/blog/next-generation-meta-training-inference-accelerator-AI-MTIA/>

Primary source; accessed by 2026-09-28

Conference records and implementation disclosures

[C01] MTIA300 ISCA2026 paper

https://aisystemcodesign.github.io/papers/MTIA300_ISCA2026.pdf

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[C02] MTIA v1, ISCA2023

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Official indexed record or related announcement; full document download unavailable.

[C03] Second generation MTIA, ISCA2025

<https://dl.acm.org/doi/10.1145/3695053.3731409>

Official indexed record or related announcement; full document download unavailable.

[C04] HCCL collective communications, 2026 preprint

<https://arxiv.org/pdf/2608.00358>

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[C05] ISSCC2019 LeCun plenary record

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Launches, ecosystem and deployment

[E01] Four MTIA chips in two years, March 2026

<https://ai.meta.com/blog/meta-mtia-scale-ai-chips-for-billions/>

Primary source; accessed by 2026-09-28

[E03] Infrastructure evolution, 2025

<https://engineering.fb.com/2025/09/29/data-infrastructure/metasp-infrastructure-evolution-and-the-advent-of-ai/>

Primary source; accessed by 2026-09-28

[E04] MSVP first video transcoding ASIC

<https://ai.meta.com/blog/meta-scalable-video-processor-MSVP/>

Indexed record or searchable official page; product details cross-checked with the associated technical document.